

Education

M.S. Computer Science	University of Utah	2025 - 2026 <i>expected</i>
B.S. Computer Science, Mathematics Minor	University of Utah	2021 - 2025
Certificate in Data Science	University of Utah	2021 - 2025

Awards & Honors

Grateful Alumni Scholarship	University of Utah	Fall 2024
Kiri Wagstaff AI/ML Scholarship	University of Utah	Fall 2024
Richard B. & Brenda R. Brown Endowed Scholarship	University of Utah	Fall 2024
College of Engineering Departmental Scholarship	University of Utah	Fall 2023, Spring 2024
Dean's List	University of Utah	2021 - 2025 (<i>all semesters</i>)

Experience

Computer Systems Teaching Assistant	University of Utah	January 2026 - May 2026
- Helped over 175 students by managing Piazza discussions, grading assignments, and provided detailed feedback on student submissions. - Ran labs and held help hours for students to address questions and clarify concepts.		
Software Practice II Teaching Assistant	University of Utah	August 2023 - May 2025
- Supported over 200 students by managing Piazza discussions, grading assignments, and provided detailed feedback on student submissions. - Ran labs and held help hours for students to address questions and clarify concepts.		
GREAT Summer Camp Instructor (4th-6th Grade)	University of Utah	June 2024 - July 2024
- Taught robotics concepts to elementary students with hands-on activities. - Supervised a structured learning environment for children.		

Projects

Goldshelf - Fullstack Ranking System	TypeScript ↗	July 2024 - Present
- Developed a full-stack web application for ranking and reviewing content of any user defined type. Hosted on Cloudflare with D1/R2 database/storage. - Custom self repairing binary search insertion to maintain a sorted list of user defined content.		
Full Stack Photo Sharing App (Capstone Project) - BeThere	C#, React	August 2024 – May 2025
- Developed a backend server with a modular REST API to handle photo uploads, user management, and clustering based on geolocation data. - Integrated PostgreSQL for robust data storage with LINQ, and built access control linking users to clusters for selective album visibility by location.		
Full JPL Compiler	C++, Rust	January 2025 – May 2025
- Developed a full compiler for the JPL programming language. - Implemented the full pipeline (lexing, parsing, semantic analysis, optimization, codegen) including type checking, scope management, and code optimizations.		
Path of Memories Gamejam	C# with Unity ↗	January 2023
- Built a 2D platformer featuring a robust dialogue system, player progression tracking, and character-driven interactions/level design. - Added diverse player abilities including wall climbing, double jumping, and dashing.		
Circuit Simulator QT Application	C++	April 2023
- Designed an educational circuit-logic game with an intuitive, interactive UI and user-customizable gates. - Implemented save functionality using JSON for persistent user data.		
LMS Website	C#	April 2023
- Developed a learning management system leveraging a MariaDB backend. - Ensured seamless front-end/back-end integration using .NET technologies, including user authentication and course management.		
Sprite Editor QT Application	C++	March 2023
- Created a comprehensive sprite editor with detailed UI and user interactions. - Supported multiple image formats/export options and implemented advanced drawing tools with color manipulation features.		
Snake Network Game	C#	October 2022 – December 2022
- Built a networked snake game with distinct server and client implementations. - Managed multiplayer functionality with real-time synchronization.		

Misc

Advent of Code 2025	Rust ↗	December 2025 - January 2026
- Solved 22 challenges focusing on dynamic programming, graph algorithms, and geometric problems. - Improved Rust proficiency while implementing efficient data structures/algorithms to optimize solution performance.		
Project Euler	C++, Python	August 2020 – December 2020
- Tackled 59 mathematics and computation-intensive problems. - Employed efficient algorithms for large-scale numerical challenges, strengthening mathematical reasoning and coding proficiency.		

Technology

Experienced
C++, Rust, C, C#, SQL, VSCode, Qt, Docker, Linux, MacOS
Proficient
Python, Java, Latex, Typst, R, Git, MS Visual Studio, MS Office, XCode